

The Taylor tree for $d = 2$ with equal starting exponents

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Abstract

The paper's Theorem (Taylor Tree) for $d = 2$ and equal starting exponents $(h_1 + 1)/k_1 = (h_2 + 1)/k_2 = p$, in the normalisation $N = \sqrt{n}$: for analytic ξ, η with weighted-summable coefficient arrays at a radius $\rho > b$ and for EVERY T ,

$$Z(N) - \sum_{\alpha \in \Lambda(h, k), \alpha < 2T} N^{-\alpha} (A_\alpha \log N + B_\alpha) = O(N^{-2T} (1 + \log N)),$$

with A_α, B_α the canonical coefficients of unit 121, defined once and independent of T (`twoD_taylor_tree_equal`). The truncation orders $M_\ell = \lceil k_\ell R \rceil$, $R = \max(1, 2T - p)$, satisfy both compatibility inequalities and push the remainder below N^{-2T} ; the pole set below $2T$ is characterised without reference to the cutoff (`mem_polesBelow_iff`) and retained poles $\geq 2T$ are discarded into the remainder. Zero `sorry/axiom`.

1 The things formalised

```
noncomputable def cutoffR (p T : ) : := max 1 (2 * T - p)
noncomputable def cutoffM (k : ) (p T : ) : := (k : ) * cutoffR p T
theorem cutoffM_bounds (k : ) (hk : 0 < k) (p T : ) :
  ((cutoffM k p T : ) - 1) / k < cutoffR p T
  cutoffR p T (cutoffM k p T : ) / k
theorem cutoff_ge (k k : ) (hk : 0 < k) (hk : 0 < k) (p T : ) :
  2 * T - p + min ((cutoffM k p T : ) / k) ((cutoffM k p T : ) / k)
noncomputable def polesBelow (h h k k : ) (p T : ) : Finset :=
  (poleSet h h k k (cutoffM k p T) (cutoffM k p T)).filter (· < 2 * T)
theorem mem_polesBelow_iff (h h k k : ) (hk : 0 < k) (hk : 0 < k) (p T : )
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p) ( : ) :
  polesBelow h h k k p T
  < 2 * T (( i, uExp h k i = ) j, vExp h k j = )
theorem discard_isBig0 (T A B : ) (h : 2 * T ) :
  (fun N : => N ^ (-) * (A * Real.log N + B)) =0[atTop]
  fun N : => N ^ (-(2 * T)) * (1 + Real.log N)
theorem twoD_taylor_tree_equal ( b p : ) (h h k k : ) (h : 0 < ) (hb : 0 < b)
  (hb : b < ) (hk : 0 < k) (hk : 0 < k) (hp : ((h : ) + 1) / k = p)
  (hp : ((h : ) + 1) / k = p) (x y : * → ) (hx : WSummable x)
  (hy : WSummable y) (T : ) :
  (fun N : => twoDAmp b N h h k k (anaAmp (ampCoeff x y) b)
    - polesBelow h h k k p T,
    N ^ (-) * (canonA h h k k (fun i j s => ampCoeff x y (i, j) s)
      * Real.log N
      + canonB b h h k k (anaFaceU (ampCoeff x y) b)
      (anaFaceV (ampCoeff x y) b) (fun i j s => ampCoeff x y (i, j) s) ))
  =0[atTop] fun N : => N ^ (-(2 * T)) * (1 + Real.log N)
```

2 Mathematics

With $R = \max(1, 2T - p)$ and $M_\ell = \lceil k_\ell R \rceil$ we have $k_\ell R \leq M_\ell < k_\ell R + 1$, hence $(M_\ell - 1)/k_\ell < R \leq M_\ell/k_\ell$. Both compatibility inequalities $(M_1 - 1)/k_1 < R \leq M_2/k_2$ (and symmetrically) follow, and $p + \min_\ell M_\ell/k_\ell \geq p + R \geq 2T$, so the canonical cutoff theorem of unit 122 gives the expansion over Λ_M with remainder $O(N^{-2T}(1 + \log N))$. Splitting Λ_M by the predicate $\alpha < 2T$, the discarded terms have $N^{-\alpha} \leq N^{-2T}$ and $|A \log N + B| \leq (|A| + |B|)(1 + \log N)$ for $N \geq 1$. Conversely every pole $\alpha_i < 2T$ has $i < k_1(2T - p) \leq k_1 R \leq M_1$, so nothing below $2T$ is missed: the retained set is exactly $\{\alpha \in \Lambda(h, k) : \alpha < 2T\}$, independent of how the cutoff was chosen.

3 Paper-facing meaning

This is `thm:TaylorTree` for $d = 2$ under the equal-exponent hypothesis, with $Z(\beta, n; \xi, \eta) = Z(N)$ at $N = \sqrt{n}$, exponent labels $\alpha = 2\mu$, and $P_\mu(X) = \frac{1}{2}A_\alpha X + B_\alpha$ (degree $\leq d - 1 = 1$). Compared with the paper’s statement: (i) the hypothesis is weighted summability of the coefficient arrays of ξ, η at some $\rho > b$, which holds for functions holomorphic on $|z_i| < R$, $R > b$; (ii) the expansion is stated with a rigorous remainder for every T rather than as a formal asymptotic series; (iii) the coefficients are canonical finite parts and Gaussian s -moments of the exponential coefficient array, and their identification with the paper’s “absolutely convergent series in derivatives of ξ, η ” is the finite-part/series formula of Astra #8 rank 4, still to do. Unequal starting exponents (rank 3) are the other outstanding piece of the $d = 2$ theorem.

4 Lean notes

- `div_lt_iff` needs the fraction isolated: derive $i/k_1 < 2T - p$ by `linarith` first, then rewrite; `nlinarith` is not needed once the product $k_1(2T - p) \leq k_1 R$ is stated via `mul_le_mul_of_nonneg_right`.
- `le_of_not_lt` is gone; use `not_lt.1`.
- `Finset.sum_filter_add_sum_filter_not s p` splits a sum by a predicate; with `set` names for the pole set and the coefficient functions the resulting identity closes by `ring`.
- `IsBig0.sum` produces a Pi-sum of functions; pass `A` explicitly and finish with `congr_left` and `simp` only [`Finset.sum_apply`].
- `Real.rpow_le_rpow_of_exponent_le (hN : 1 ≤ N)` compares powers of the same base ≥ 1 .